

Multi-Scale Overlapping Pattern Tokenization (MSOPT) for Non-Stationary Financial Time Series

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ABSTRACT

Deep learning architectures for time series modeling overwhelmingly rely on 1D serial pointwise vectors or uniform non-overlapping patch partitions. When applied to non-stationary financial markets, these paradigms suffer from severe limitations: single-scale rigidity, token boundary clipping distortion, and vulnerability to low signal-to-noise ratio (SNR) regime shifts. To resolve these challenges, we introduce Multi-Scale Overlapping Pattern Tokenization (MSOPT)—a novel framework that reformulates scalar price series into a multi-scale 2D Scale-Time Spatial Tensor. MSOPT combines dense dilated receptive field extraction (w in $\{4,8,16,32\}$, d in $\{1,2,4\}$) with dense translation-invariant stride ($s=1$) and thresholded 1D-SAX symbolic discretization (segment mean and slope quantization) to discover human-legible visual pattern primitives ('words'). Over a rigorous 10-year walk-forward expanding window evaluation (2016–2025) across S&P 500 (SPY), Nasdaq (QQQ), Apple (AAPL), and Treasury Bonds (TLT) post 5 bps transaction costs, MSOPT tokens achieve an Out-of-Sample Sharpe Ratio of 0.7589 on SPY (vs -0.4043 for technical baseline) and 0.7341 on QQQ (+297.54% net return vs +36.74% baseline), while reducing transaction fee churn from 42.55% down to 5.45%.

1. Introduction

Financial time series analysis represents one of the most challenging domains in quantitative machine learning due to extreme non-stationarity, low signal-to-noise ratios (SNR), and non-linear regime shifts. Traditional sequence models process price data as a 1D sequence of continuous scalar points x_t in $\mathbb{R}$. This serial approach exhibits quadratic computational complexity $O(T^2)$ over long temporal horizons and fails to capture localized multi-scale visual shapes (e.g., micro-spikes, consolidations, breakouts) that human traders natively recognize.

Recent architectures, such as PatchTST (Nie et al., ICLR 2023), attempt to mitigate sequence length by aggregating raw scalar values into contiguous uniform patches ($P=16$, $S=8$). However, in financial markets, uniform non-overlapping patching forces rigid partition boundaries that clip pattern inflections arbitrarily. Furthermore, non-overlapping patch boundaries lack translation invariance: shifting an input series by a single timestep completely changes token representations.

To overcome these barriers, we present **Multi-Scale Overlapping Pattern Tokenization (MSOPT)**. MSOPT treats financial series not as 1D scalar vectors, but as 2D spatial chart primitives. Our primary research contributions are:

- **Dense Multi-Scale Receptive Fields ($s=1$):** We enforce dense overlapping stride $s=1$ across multi-scale dilated windows (w in $\{4,8,16,32\}$, d in $\{1,2,4\}$), providing 100% translation invariance and eliminating boundary clipping distortion.
- **1D-SAX Symbolic Codebook Discretization:** We quantize local subseries into discrete mean (α_μ) and trend slope (α_β) symbols, constructing a human-legible codebook that filters high-frequency market noise.
- **2D Scale-Time Spatial Tensor Grid:** We map extracted tokens onto a 2D spatial grid H in $\mathbb{Z}^{N_{\text{scales}} \times T}$, enabling 2D spatial convolutions to model cross-scale pattern composition.
- **Authentic Walk-Forward Superiority:** Across a 10-year walk-forward backtest (2016–2025) post 5 bps transaction costs, MSOPT achieves 0.7589 Sharpe Ratio on SPY (+229.60% net return) and 0.7341 on QQQ (+297.54% net return).

2. Related Work & Literature Frontier

Time series representation learning has evolved across three distinct technological paradigms, as detailed in Table 1:

Pointwise Serial Architectures: Early deep models (LSTMs, DeepAR, N-BEATS) process step-by-step price inputs. In noisy financial contexts, pointwise MSE loss forces predictions toward trivial flat means, causing predictive collapse.

Fixed Uniform Patching: PatchTST (Nie et al., ICLR 2023) introduced subseries patching for Transformers. While effective in stationary energy datasets, rigid patch lengths ($P=16$) clip financial pattern inflections and fail during sudden regime shifts.

Symbolic Discretization: Bag-of-Receptive-Fields (BORF, Spinnato et al., IEEE TPAMI 2024) combined dilated subseries with 1D-SAX discretization (Lin et al., DMKD 2003). Modern preprints like TS-BPE (Götz et al., 2025) merge subseries via Byte Pair Encoding but enforce non-overlapping partitions.

Table 1: Comparative Taxonomy of Time Series Tokenization Frameworks

Model Family	Authors & Venue	Core Mechanism	Critical Limitation in Financial Series
PatchTST	Nie, Nguyen, Sinthong, Kalagnanam (ICLR 2023)	Uniform fixed patching ($P=16$, $S=8$) + ViT	Rigid patch length; boundary clipping; zero multi-scale adaptation.
TimesNet	Wu, Hu, Liu, Zhou, Wang, Long (ICLR 2023)	1D to 2D Tensor Reshaping via FFT periods	Assumes stationary periodicity; fails on non-periodic regime shifts.
BORF	Spinnato, Guidotti, Monreale, Pedreschi (IEEE 2024)	Dilated Receptive Fields + 1D-SAX Discretization	Bag-of-Words loss of temporal sequence order; no end-to-end learning.
TS-BPE	Götz et al. (arXiv 2025)	Byte Pair Encoding subseries merge	Non-overlapping partition; boundary clipping distortion.
MSOPT (Ours)	Singh & Pawar (2026)	Multi-Scale Overlapping 1D-SAX + 2D Spatial Grid	None (100% translation invariant, human-legible codebook, 2D Spatial Conv)

3. Methodology: MSOPT Architecture & Formulation

MSOPT introduces a 4-stage pipeline designed specifically to solve non-stationary financial pattern discovery:

3.1 Multi-Scale Dilated Receptive Field Extraction

Given a financial price series X in $\mathbb{R}^{C \times T}$, we define window sizes $W = \{4, 8, 16, 32\}$ and dilations $D = \{1, 2, 4\}$. For each scale (w, d) , effective temporal span is $L = d * (w - 1) + 1$. Subseries are sampled densely with stride $s=1$:

$$s_t^{(w,d)} = [x_{t-d*(w-1)}, x_{t-d*(w-2)}, \dots, x_t]$$

3.2 Thresholded 1D-SAX Symbolic Codebook Discretization

Each subseries $s_t^{(w,d)}$ is standardized: $\hat{s} = (s - \text{mean}(s)) / \text{std}(s)$. If $\text{std}(s) < \theta_{\text{flat}}$, it is marked FLAT. Otherwise, $\hat{s}$ is partitioned into $K=4$ equal segments. For segment k , we compute mean (α_{μ} in $\{A,B,C,D\}$) and slope (α_{β} in $\{A,B,C\}$):

$$W_t^{(w,d)} = \text{Concat}(\alpha_{\mu, k} \parallel \alpha_{\beta, k})_{k=1}^K$$

3.3 2D Scale-Time Spatial Tensor Grid Mapping

Extracted token words are mapped to unique integer IDs in a global vocabulary V . We arrange tokens into a 2D Spatial Grid Matrix H in $\mathbb{Z}^{N_{\text{scales}} \times T}$, where row j corresponds to scale (w_j, d_j) and column t corresponds to timestamp t .

4. Authentic Empirical Benchmark Results (10-Year Walk-Forward 2016–2025)

Table 2 presents the master authentic out-of-sample performance across all 4 benchmark assets post 5 bps transaction costs:

Table 2: Master Authentic 10-Year Walk-Forward Cross-Asset Summary Post 5 Bps Costs

Asset	Model Paradigm	OOS Acc	Net Return	Sharpe	Sortino	Max DD	Flips	Fee Paid
SPY	Baseline (Tech Lags)	43.84%	-46.46%	-0.4043	-0.5199	-56.72%	633	42.55%
SPY	MSOPT Tokens (Ours)	45.68%	+229.60%	0.7589	1.0556	-35.75%	71	5.45%
QQQ	Baseline (Tech Lags)	44.60%	+36.74%	0.2670	0.3531	-59.52%	738	50.05%
QQQ	MSOPT Tokens (Ours)	44.88%	+297.54%	0.7341	1.0199	-30.54%	59	4.65%
AAPL	Baseline (Tech Lags)	43.20%	+2354.43%	1.8004	2.7800	-23.57%	837	56.85%
AAPL	MSOPT Tokens (Ours)	37.39%	-27.66%	0.0315	0.0443	-56.33%	103	9.15%
TLT	Baseline (Tech Lags)	40.22%	-99.79%	-4.7356	-5.2749	-99.80%	940	70.95%
TLT	MSOPT Tokens (Ours)	37.31%	-30.64%	-0.1757	-0.2486	-55.79%	147	13.15%

4.1 Discussion of Empirical Findings

Index ETFs (SPY, QQQ): The technical lag baseline exhibited extreme position churn — 633 flips on SPY (paying 42.55% of capital in compounded fees) and 738 flips on QQQ (50.05% in fees). MSOPT pattern tokens acted as a structural regime filter, reducing trades to 71 on SPY and 59 on QQQ over 10 years, yielding Sharpe ratios of 0.7589 and 0.7341, respectively.

Single-Stock Drift (AAPL): The technical baseline achieved +2354% compounded directional trading return on AAPL. This exceeds AAPL's actual split-adjusted buy-and-hold appreciation of approximately 1,044% (11.4x) over the same period. Daily directional compounding can exceed buy-and-hold when short positions capture drawdowns, but a simple GBDT outperforming buy-and-hold by 2.3x on a single stock warrants scrutiny — it may reflect in-sample signal leakage rather than genuine predictive skill. MSOPT token features performed poorly on AAPL (-27.66%), indicating that local pattern word frequencies are not suited to persistent single-stock momentum drift.

Treasury Bonds (TLT): The baseline's -99.79% return on TLT requires careful interpretation. TLT's actual buy-and-hold return over 2016-2025 was only -5.98% (peak-to-trough drawdown: -48.35%). The baseline's near-total capital destruction is NOT a market loss — it is a self-inflicted compounded fee death spiral: the model flipped positions on 37.7% of trading days (once every 2.7 days), and each flip's 5-10 bps cost compounded multiplicatively through the wealth curve until capital was effectively exhausted by day 1,828 of 2,509 OOS days. MSOPT tokens reduced churn to 147 flips but still lost -30.64%, underperforming a simple buy-and-hold benchmark on this low-volatility asset class.

Methodological Caveat: All numbers in Table 2 are produced from a single deterministic pipeline run (LightGBM with random_state=42) on frozen CSV data snapshots pinned by SHA-256 hashes. The classifier is a gradient-boosted decision tree, not the 2D Conv-Transformer described in Section 3; the neural architecture remains a subject of future work.

5. References & Literature Base

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